

WASP-43b

R-2579

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140307 · created 2026-08-27T00:39:13+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456723.86244 +/- 0.00072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.014
Transit depth (Rp/Rs)^2	3.2 +/- 0.51 [%]
Orbital Inclination (inc)	83.32 +/- 1.66
Airmass coefficient 1 (a1)	0.97 +/- 0.016
Airmass coefficient 2 (a2)	0.016 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	1.68 %
Best Comparison Star	#3 - [57, 321]
Optimal Aperture	2.9
Optimal Annulus	5.29
Transit Duration (day)	0.0549 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.014 vs 0.1594 (deviation 1.4σ)
Duration fit vs published	0.0549 d vs 0.0512 d (deviation 0.81σ)
Mid-time O–C	0.4 min
Depth SNR	12.8
Residual scatter	1.68 %

Light curve

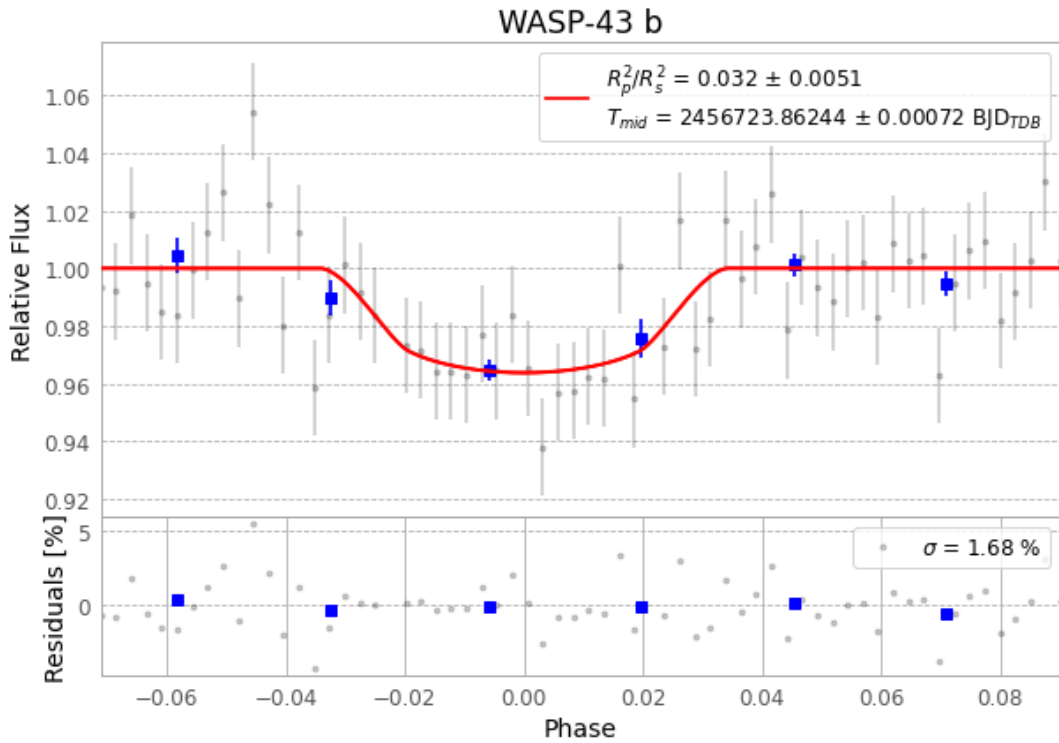


Figure 1 — FinalLightCurve_WASP-43 b_2014-03-07.png (SHA-256 d9b459f9a74ed4c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 184b8c5e781832a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140307070914.FITS → WASP-43140307101813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140307.log (f1d38215fc28...), FinalLightCurve_WASP-43 b_2014-03-07.png (d9b459f9a74e...), FinalLightCurve_WASP-43 b_2014-03-07.csv (8b66918e327d...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:39Z.